Lecture 15

- Test for gradient fields (16.3)
- Green's Theorem (16.4)

$\vec{F} = \langle P, Q, R \rangle$

If $\vec{F} = \nabla f = \langle f_x, f_y, f_z \rangle$, then

$$P_y = f_{xy} = f_{yx} = Q_x$$
$$Q_z = f_{yz} = f_{zy} = R_y$$
$$R_x = f_{zx} = f_{xz} = P_z$$

<u>Theorem</u>: If $\vec{F} = \langle P, Q, R \rangle$ is defined for all (x, y, z), and

$P_y = Q_x$, $Q_z = R_y$, $R_x = P_z$

then $\vec{F} = \nabla f$ for some f.

<u>Example</u>:

$$\vec{F} = \langle \underbrace{x^3 - 3xy^2}_{P}, \underbrace{y^3 - 3x^2y}_{Q}, \underbrace{30z}_{R} \rangle$$

$$\left.\begin{aligned} P_y &= -6xy = Q_x \\ Q_z &= 0 = R_y \\ R_x &= 0 = P_z \end{aligned}\right\} \Rightarrow \vec{F} = \nabla f$$

<u>How to find f</u>

$f_x = x^3 - 3xy^2$ (1)

$f_y = y^3 - 3x^2y$ (2)

$f_z = 30z$ (3)

$\int dx$:

$$f(x,y,z) \overset{1}{=} \frac{x^4}{4} - \frac{3}{2}x^2y^2 + g(y,z)$$

$\partial/\partial y$:

$$\Rightarrow f_y \overset{2}{=} -3x^2y + g_y(y,z)$$
$$= y^3 - 3x^2y$$
$$\Rightarrow g_y = y^3$$

$\int dy$:

$$g(y,z) = \frac{y^4}{4} + h(z)$$

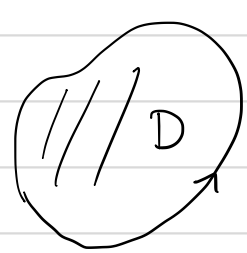
$\partial/\partial z$:

$$g_z = h'(z) \overset{3}{=} 30z$$
$$\Rightarrow h = 15z^2 + C$$
$$\Rightarrow g = \frac{1}{4}y^4 + 15z^2 + C$$

$$\Rightarrow \boxed{f = \frac{1}{4}x^4 - \frac{3}{2}x^2y^2 + \frac{1}{4}y^4 + 15z^2 + C}$$

<u>Green's theorem</u>

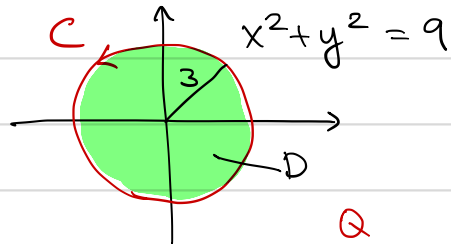
$$\boxed{\int_C P\,dx + Q\,dy = \int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA}$$



C: closed curve (positively oriented)

D: region bounded by C

Example:



$x^2+y^2=9$

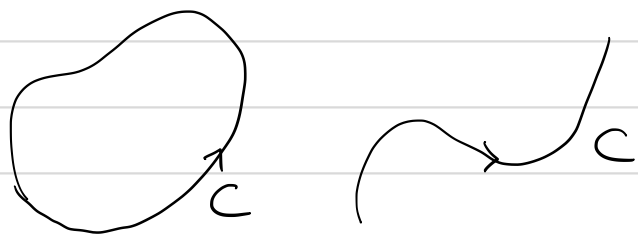
$$\int_C \overbrace{(3y-e^{\sin x})}^{P}\,dx + \overbrace{(7x+\sqrt{y^4+1})}^{Q}\,dy$$

$$= \iint_D \frac{\partial}{\partial x}(7x+\sqrt{y^4+1}) - \frac{\partial}{\partial y}(3y-e^{\sin x})\,dA$$

$$= \int_{\theta=0}^{2\pi}\int_{r=0}^{3} (7-3)\, r\,dr\,d\theta$$

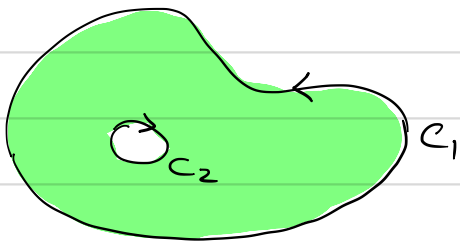
$$= 4\cdot 2\pi \cdot \int_0^3 r\,dr = 8\pi \left.\frac{r^2}{2}\right|_0^3 = 36\pi$$

What the words mean:



closed not closed

"C positively oriented" means: region D to the left of C.



Outer curves: counter-clockwise

Inner curves: clockwise

Criterion for gradient fields:

If $Q_x - P_y = 0$, then

$$\int_{\substack{\text{closed}\\ \text{loops}}} \vec{F}\cdot d\vec{r} \overset{\text{Green's}}{=} 0,$$

so $\vec{F}$ is conservative $(=\nabla f)$.